Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Engineered pluggable **distributed executor** routing pipeline nodes to **Docker** or **AWS Lambda**, backed by **S3** artifact store eliminating redundant recomputation.
- Designed stateful recovery via per-node **Redis checkpointing**; interrupted workflows **resume from the last committed step** across **distributed execution**.
- Deployed **RAG pipeline** on **AWS SageMaker** for real-time decision support at **sub-200ms p99 latency** with **RTO under 2 minutes**.
- Built **multi-agent system** via **OpenAI Agents SDK** + **MCP** tool-use handling **millions of daily transactions** at **99%+ uptime**.

QureAI | *ML Infrastructure Engineer* Mar 2026 – Present

- Improved **RAG retrieval recall 61% to 89%** on a 2K-query eval set by rebuilding chunking with **hybrid dense-sparse retrieval** and **re-ranking**.
- Built **model observability** framework tracking **drift**, latency, and **eval regressions** in production, enabling governance-aligned **rollback** workflows.
- Cut **p99 inference latency 420ms to 140ms** across **50K+ daily requests** migrating synchronous serving to async **vLLM** on **GCP Cloud Run** with autoscaling.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Built and operated **25+ production microservices** in **C++** and SQL-backed **distributed systems** for **1M+ users**, sustaining **99.9% uptime** over 4 years on **AWS** (EC2, RDS, SQS).
- Standardized API contracts and failure handling across **25+ services**, reducing escalations **60%**; drove **root cause analysis** and **post-incident reviews** to improve long-term reliability.

Projects

LangFetch `tinyurl.com/LangFetch`

- **LangGraph** agent decomposes multi-table questions into dependency-ordered **SQL subqueries** with intermediate result passing; deployable as **MCP**-compatible server.

Distributed KV Store / VPC Control Plane Simulator `github.com/ganeshhgupta/distributed-kv-store`

- **Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator modeling subnet routing and security groups.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system diagnosing **production incidents** via tool-use chains; routes alerts to specialized sub-agents and synthesizes **root cause reports**, reducing **MTTR**.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Jul 2016 – Jul 2020

Skills

Languages: Python, TypeScript, Java, Go, SQL

Data Infra: RAG Pipelines, Multi-Table SQL Decomposition, Data Retrieval, MCP-Compatible Services

AI/ML: OpenAI Agents SDK, LangGraph, Model Observability, AWS Bedrock

Infra: AWS, GCP, Docker, Kubernetes, PostgreSQL, Redis

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer